

Structural basis of psychedelic effect: a DiPT-focused computational program

Master project report: analyses completed and planned

This report consolidates every analysis in the project, from the original three-psychedelic comparison through the DiPT deep-dive to the two execution-ready experimental designs. It records what was done, what was found, and what remains to be run.

1. The question, and how it narrowed

The project began with a broad aim: **understand the structural basis for why different psychedelics produce different subjective effects**. The initial comparison set LSD, DiPT, and mescaline against one another (three chemotypes: ergoline, tryptamine, phenethylamine, with distinct phenomenology) and asked whether their receptor-binding profiles, brain distribution, and pocket-level docking could explain the differences.

That comparison converged on one compound as the genuine anomaly. LSD and mescaline are "visual" psychedelics whose effects track their 5-HT_{2A} engagement in the expected way. **DiPT is the outlier**. Its principal reported effect is *auditory* (a downward pitch distortion, "diptogenia"), it is a weak and atypical 5-HT₂ ligand, and it sits at the end of the N,N-dialkyltryptamine series, where isopropyl branching collapses the affinity that normally drives psychedelic activity. Every subsequent phase drilled into that anomaly, and the question sharpened from "why do psychedelics differ" to "**why is DiPT auditory-specific, and where does that effect physically live?**"

The short answer the program arrived at was: not in the classical psychedelic receptors. Two independent lines, our own negative off-target screen and the only mechanistic paper on diptogenia, point to a peripheral, cochlear locus. The computational work below establishes that conclusion and then tests how far structure-based prediction can chase it.

2. Analyses completed

2.1 Three-psychedelic structure–effect comparison (binding, ADME, spatial, docking)

The foundational phase built a four-track profile of LSD, DiPT, and mescaline.

Binding. Measured affinities pulled from ChEMBL across the serotonin-receptor panel, assembled into an affinity fingerprint. The headline contrast: LSD binds 5-HT_{2A} tightly (pK_i 8.46) and broadly; mescaline is a moderate 5-HT_{2A} binder (pK_i 6.26); DiPT registers a measured 5-HT_{1A} affinity (pK_i 6.92) but has **no measured 5-HT_{2A}/2C value at all**, a coverage gap that recurs throughout the project and was repeatedly reframed as *absence of data, not absence of binding*.

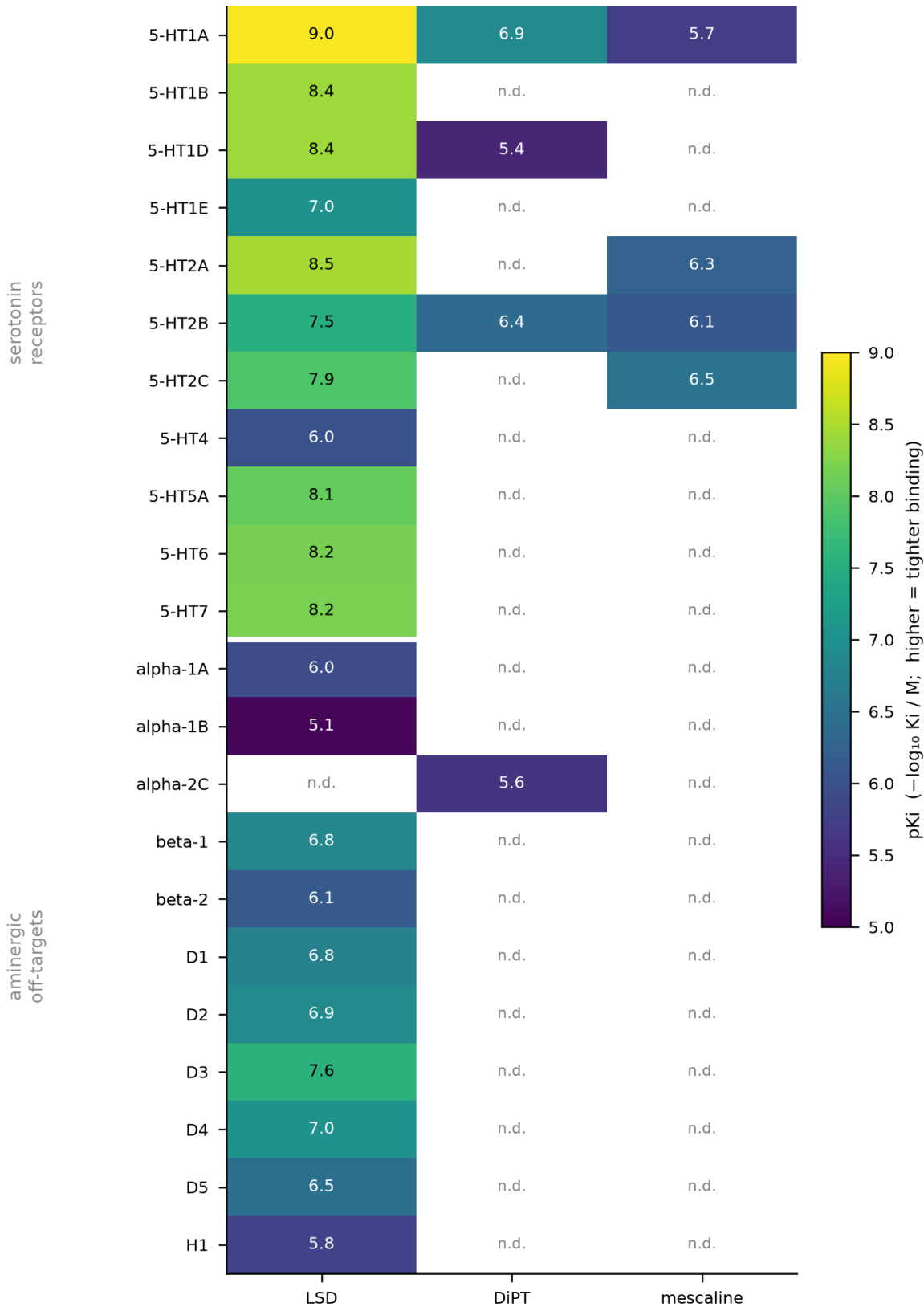
ADME / CNS penetrance. Computed physicochemical descriptors and CNS-MPO scores as a proxy for brain access (true regional drug concentrations being unavailable). LSD and mescaline score as clean CNS-penetrant molecules (MPO 5.6, 5.1); DiPT is lower (3.42), driven by its lipophilicity and basic tertiary amine.

Spatial. Receptor-expression maps (Human Protein Atlas / brain expression) stood in for regional drug distribution, showing where each target receptor is enriched across brain regions.

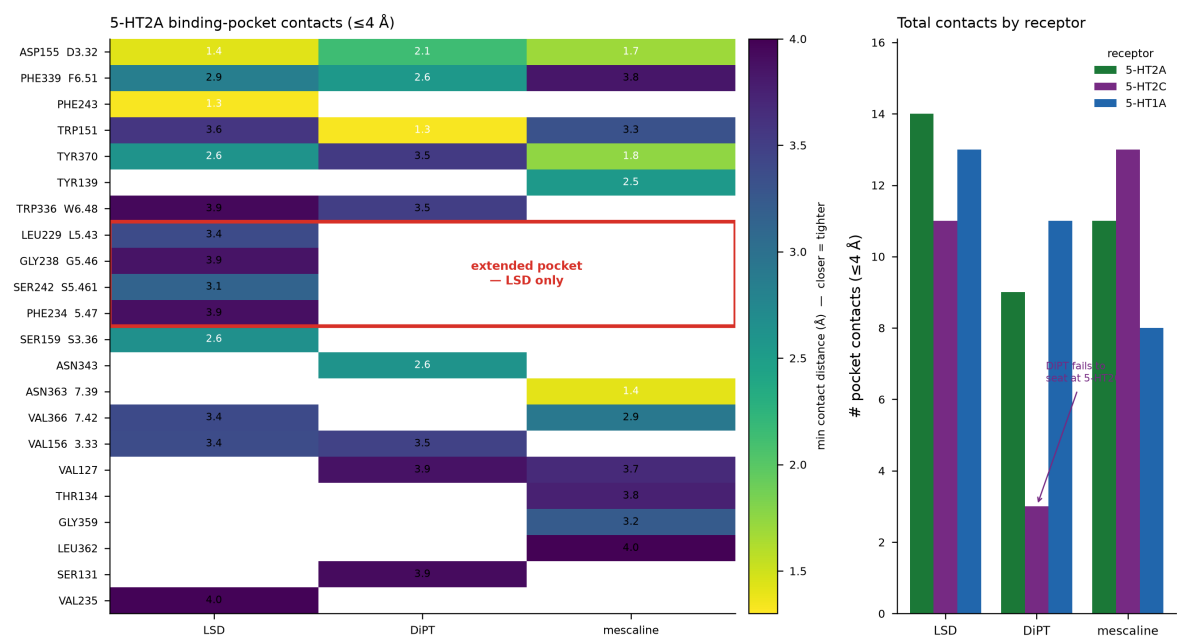
Docking. Blind DiffDock docking into three active-state GPCR structures (5-HT_{2A} = 7WC4, 5-HT_{2C} = 6BQG, 5-HT_{1A} = 7E2X), scored by orthosteric engagement and the conserved amine–D3.32 salt bridge. LSD seats deep and makes the most pocket contacts; DiPT's seating is weaker and, at 5-HT_{2C}, the amine sits 14.3 Å from D3.32, failing to engage the canonical anchor.

Figures:

LSD binds broadly and tightly; mescaline is a narrow weak 5-HT2 agonist; DiPT data are sparse



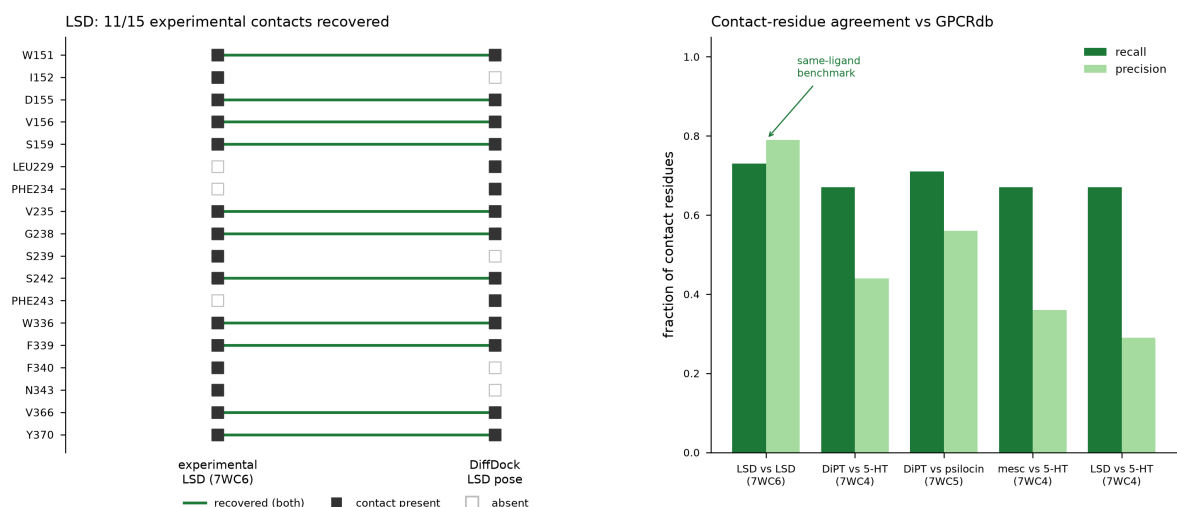
Only LSD reaches the 5-HT2A extended binding pocket — the structural correlate of its potency



2.2 Validation against GPCRdb

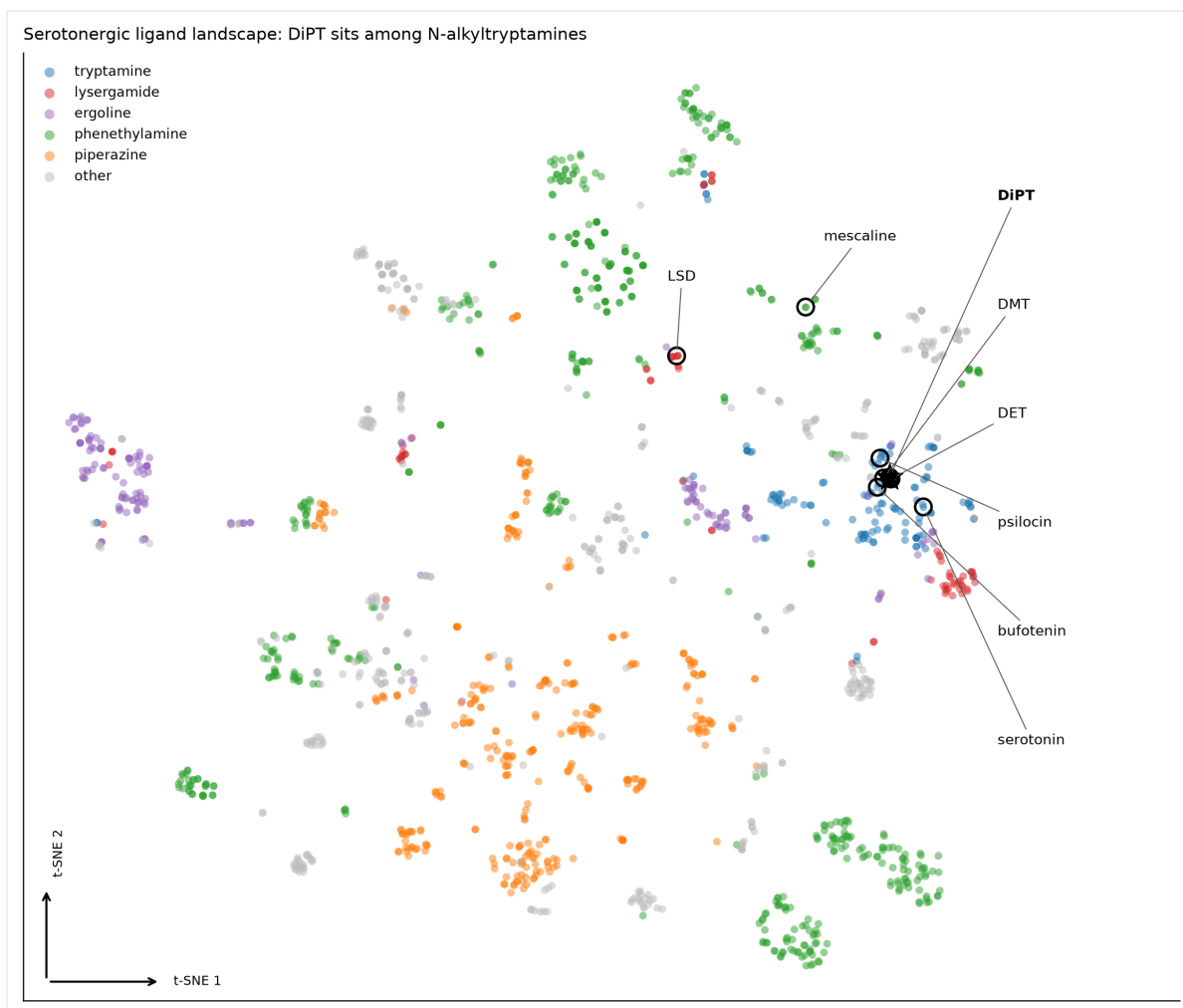
To check the docking against experiment, LSD's docked 5-HT2A contacts were compared to the crystallographic contact set (PDB 7WC6) matched by residue sequence number. **Precision 79%, recall 73%:** the docking recovers most true contacts. A residue-level audit corrected two docking false-positives (L229, F234) and confirmed V235 as a genuine hydrophobic contact. The canonical extended-pocket set was fixed at four residues (L229, G238, S242, F234).

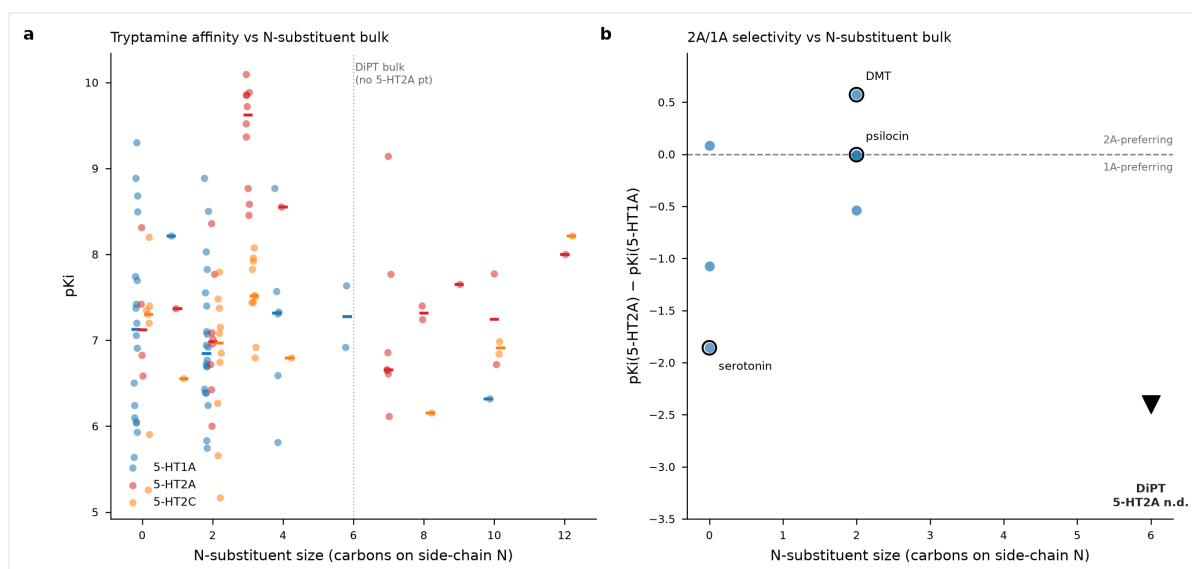
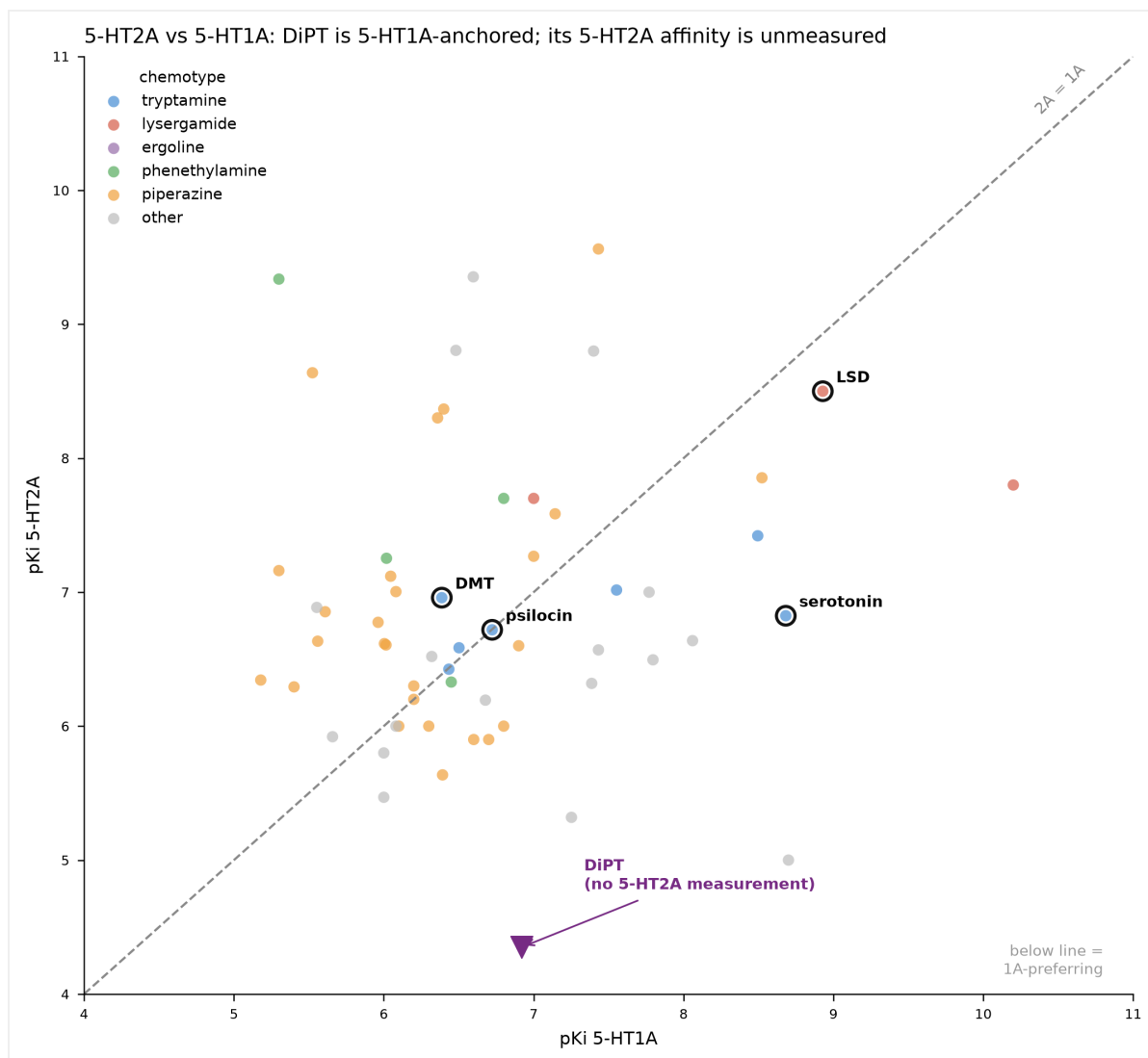
Docking poses validated against GPCRdb experimental 5-HT2A structures



2.3 Expanded chemical landscape (1,667 compounds)

To place DiPT among its measured peers, a broad ChEMBL/BindingDB scrape assembled **1,667 compounds** (tryptamines, phenethylamines, lysergamides, ergolines, piperazines) with measured 5-HT_{1A}/2A/2C affinities and computed descriptors. Chemical-similarity embedding, a 5-HT_{1A}-vs-5-HT_{2A} selectivity map, and N-alkyl SAR trends locate DiPT in a sparsely-measured tryptamine neighbourhood; only 5 tryptamines have measurements at all three receptors; this underscores how thin the data is exactly where DiPT sits.



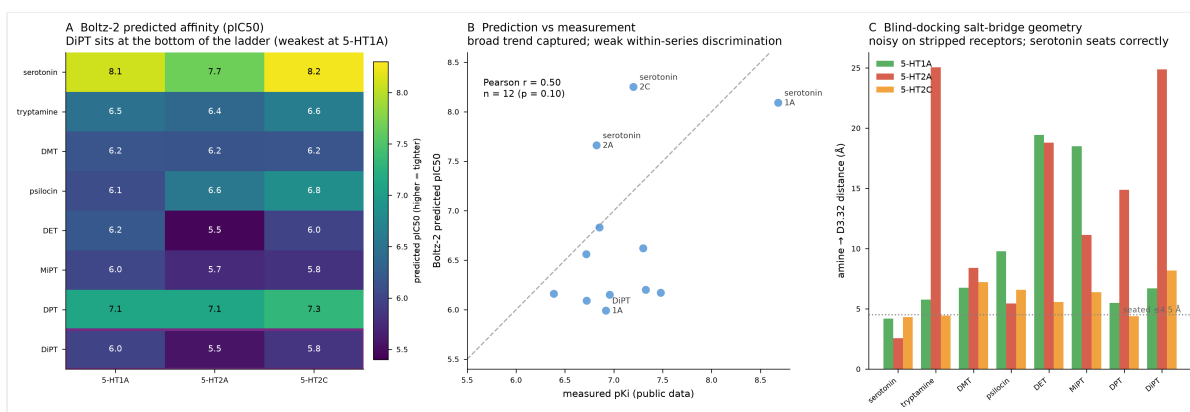


2.4 Ideation, discovery program, and literature review

A structured discovery program (2 phases, 11 steps) and a **33-reference literature review** framed DiPT's profile against the field. A companion off-target scrape found DiPT has **zero off-target measurements** in ChEMBL, establishing that the auditory-vs-visual question could not be answered by mining existing data. It is a genuine predict-then-test task.

2.5 Compute execution: DiffDock + Boltz-2 (E1/E2/E3)

The E-series ran on the user's Modal GPU account: DiffDock pose prediction and Boltz-2 co-folding of a tryptamine ladder against the three receptors (job 33f4f17a, 24 co-folds, ~\$5.74 total). **DiPT was predicted the lowest binder-probability of the ladder at all three receptors (0.62–0.69); by numeric pIC₅₀ it was weakest at 5-HT_{1A} and second-weakest at 2A/2C.** Serotonin correctly topped the ranking (pIC₅₀ ~7.7–8.3, prob ~0.99), validating the affinity head's direction. Retrospective validation on 12 measured pairs gave Pearson $r = 0.50$ — trend captured, within-series discrimination weak.



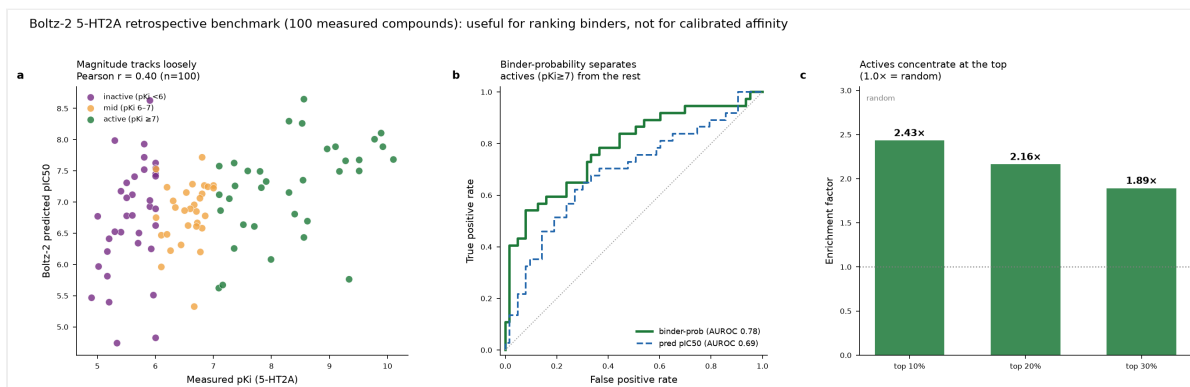
2.6 Boltz-2 benchmark and calibration (100-compound 5-HT_{2A} retrospective)

Because the numeric affinities were not obviously trustworthy, a proper retrospective benchmark was run: **100 compounds with measured 5-HT_{2A} pK_i, co-folded locally on the user's Apple-Silicon Mac** (after a 12-compound test surfaced and fixed an MSA null-byte bug that had silently produced zero predictions). Result, and this is the load-bearing calibration for the whole program:

- **Boltz-2 binder-probability is a trustworthy classifier: AUROC 0.78, BEDROC($\alpha=20$) 0.89, enrichment 2.4 $\times$ in the top decile (9/10 true actives).**

- **Numeric pIC₅₀ is NOT calibrated: AUROC 0.69, Pearson r = 0.40.**

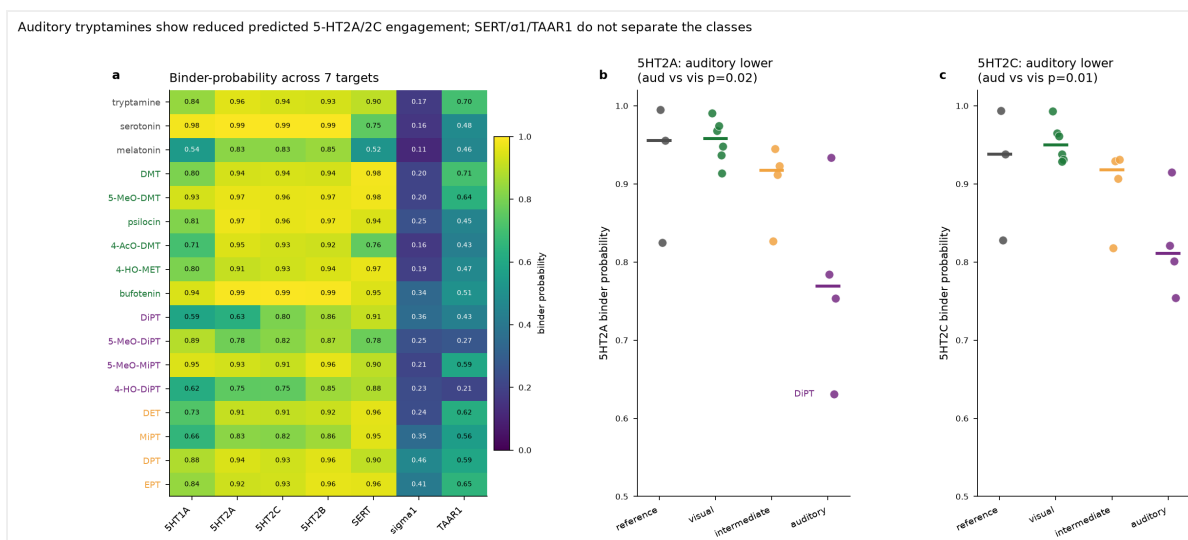
So downstream, binder-probability is read as a ranking signal and the numeric affinity is not over-interpreted. (Caveat: the "inactive" band was weak binders, not true decoys, and the 37% active fraction is high, so the absolute AUROC is optimistic.)



2.7 Phase 2 — auditory-vs-visual off-target panel ($17 \times 7 = 119$ co-folds)

The central predict-then-test experiment. Seventeen tryptamines grouped by phenomenology (4 auditory, 6 visual, 4 intermediate, 3 reference) were co-folded against 7 targets: the three serotonin receptors plus 5-HT_{2B}, SERT, sigma-1, and TAAR1. The question: does any target separate auditory from visual compounds?

Result: the only separation falls at the shared 5-HT₂ receptors, and it is the wrong direction for a distinct auditory target. Auditory tryptamines bind 5-HT_{2A} and 5-HT_{2C} *more weakly* than visual ones (5-HT_{2A} binder-prob 0.78 vs 0.96, $p=0.02$; 5-HT_{2C} 0.82 vs 0.95, $p=0.01$), with no off-target (SERT, sigma-1, TAAR1) singling out the auditory class. DiPT is the extreme case (5-HT_{2A} prob 0.63, lowest in panel). **There is no evidence for a distinct "auditory receptor" among the screened targets.**



2.8 Novelty literature review

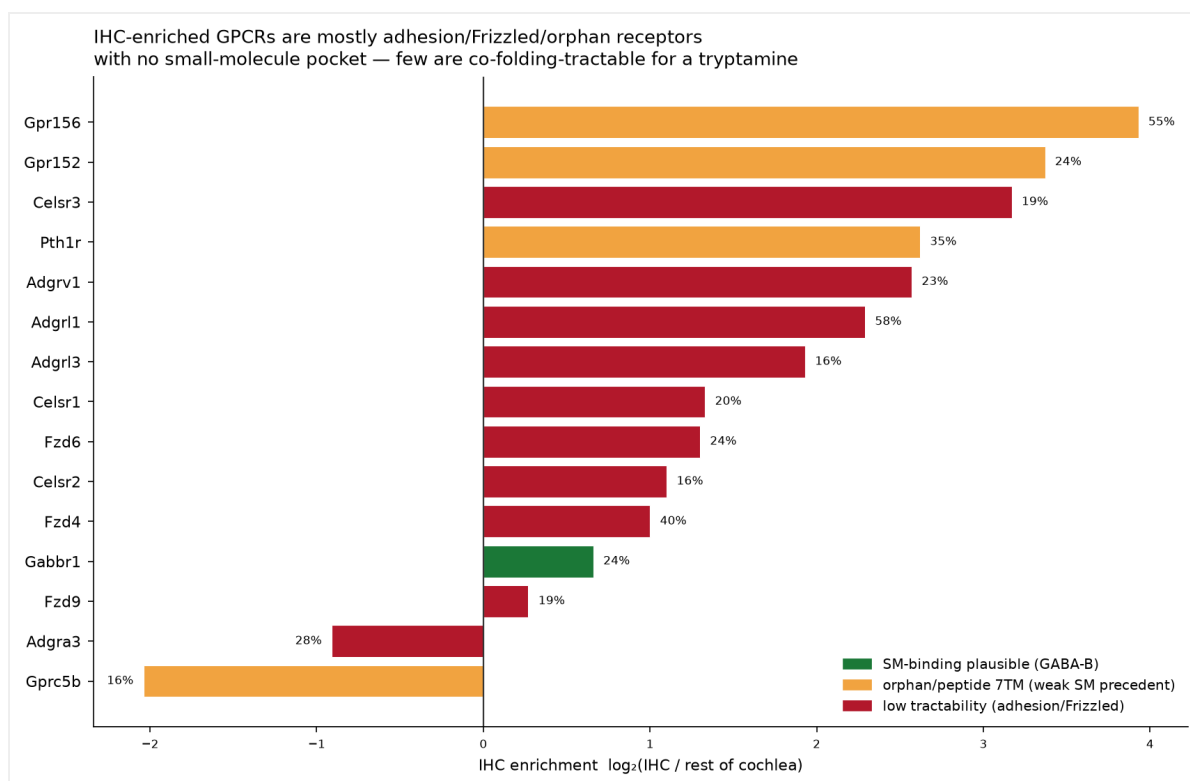
A 12-reference, DOI-verified review (none retracted) placed the Phase 2 result in context. **The affinity trend itself is not new:** McKenna 1990 and Carbonaro 2014 established that N,N-diisopropyl bulk lowers 5-HT_{2A} affinity. The novelty is narrow and methodological: the predictive multi-target framing plus a clean *negative* off-target result against a genuine data gap. The mechanism remains open, and the one paper proposing a mechanism (Hurst & Bajo 2026) places it in the cochlea.

2.9 Cochlear pivot and the IHC-GPCR feasibility screen (executed)

Reading the Phase 2 negative against Hurst & Bajo 2026 reframed the whole question: diptogenia is proposed to be a **peripheral, cochlear** phenomenon, physiologically dissociable from the emotional/psychedelic effects, with DiPT binding 5-HT_{2A} with negligible affinity. Both lines agree the auditory locus is not in the classical psychedelic receptors (the program had been screening the wrong organ).

The feasibility of a cochlear screen was then tested directly. **CELLxGENE Census carries no mammalian cochlear sensory-epithelium data**, but GEO does: Kolla 2020 (GSE137299), a P7 mouse organ-of-Corti scRNA-seq series. The matrix (21,500 genes $\times$ 3,376 cells) was downloaded, clustered, and the hair-cell cluster sub-clustered into inner (n=115, Slc17a8/Otof-high) and outer (n=149, Slc26a5-high) hair cells. GPCRs enriched in inner hair cells were ranked and structure-checked.

The expression side succeeds — 15 GPCRs are reproducibly IHC-enriched (led by Gpr156, Gpr152, Celsr3, Pth1r, Adgrv1), but the druggability side is discouraging: they are overwhelmingly adhesion, Frizzled, and orphan class-C receptors with no small-molecule pocket. Only Gabbr1 (and marginally Pth1r) has any small-molecule-binding footing, and none is aminergic. So the cochlear co-folding screen is feasible but low-prior for a small-molecule method: most IHC-specific receptors are not the kind of target a tryptamine co-fold can score.



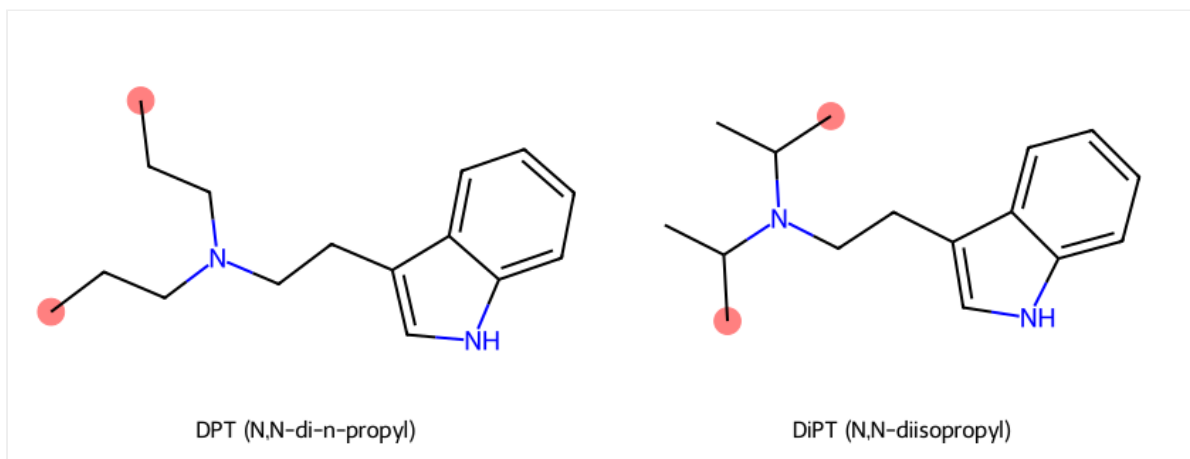
3. Analyses planned (designed, not yet run)

3.1 Cochlear IHC-GPCR co-folding screen

Co-fold the auditory-vs-visual panel against the tractable IHC-GPCR subset (Gabbr1, Pth1r, plus orphan long-shots Gpr156/Gpr152) on the same local Boltz-2 workflow, scoring on class-selectivity rather than raw binding. Framed explicitly as **exploratory and low-prior**: a null result is the likely and still-informative outcome, and would push the mechanism toward the adhesion/orphan receptors or the Cav1.3 / stria-vascularis ion machinery that this pipeline cannot adjudicate. Target list and tractability tiers are saved (`ihc_gpcr_target_shortlist.csv`).

3.2 DPT → DiPT relative binding free energy (RBF E / FEP)

The last high-value in-silico experiment on the CNS-receptor side. DPT and DiPT are **constitutional isomers** (identical $C_{16}H_{24}N_2$, 18 heavy atoms) differing only in β -branching of the N-alkyl chains, so an alchemical DPT→DiPT transformation isolates that one variable. The atom mapping is exceptionally clean (**16 of 18 heavy atoms are a shared core, only 2 soft-core atoms per side, no dummy atoms, no charge change**), the most accurate class of RBF E perturbation.



Hypothesis: $\Delta\Delta G_{\text{bind}}(\text{DPT} \rightarrow \text{DiPT})$ at 5-HT_{2A} is positive (DiPT weaker), expected order +1 to +2 kcal/mol pending measured K_i values for both compounds. **Block-billing-aware compute plan** (Modal A100-80GB, no GPU spent this pass): a **\$82 lean go/no-go edge** at 5-HT_{2A} gates a **\$617 full 5-HT_{2A} edge** (12 λ -windows, 5 ns, 3 replicates, HREX/MBAR), with an optional **+\$1,233 incremental** to extend to 5-HT_{2C} and 5-HT_{1A} ($\approx$ \$1,850 full three-receptor total). Efficiency pattern mirrors the Boltz batching: one container per leg with all λ -windows co-resident, equilibrate-once-and-checkpoint, single reused image.

3.3 Discovery-program Phases 3–5 (not started)

The broader discovery program scoped later-stage work (expanded FEP across the series, a design loop, and active-state MD) that was never begun and remains available if the program continues.

4. What the program established

1. **DiPT is a weak, atypical 5-HT₂ ligand, which is solved biology.**
Reproducing the tryptamine affinity ranking recovers known SAR (McKenna 1990, Carbonaro 2014); it is not where the discovery is.
2. **The auditory effect does not live in the classical psychedelic receptors.** Our Phase 2 negative and Hurst & Bajo 2026 agree independently. The locus is most likely peripheral (cochlear).
3. **The cochlear hypothesis is feasible to probe but structurally hard for a small-molecule method.** IHC-specific GPCRs are dominated by adhesion/orphan/Frizzled receptors with no small-molecule pocket.
4. **Boltz-2 binder-probability is a usable classifier (AUROC 0.78); its numeric affinity is not calibrated (r=0.40).** This calibration bounds every co-folding conclusion in the project.
5. **The honest computational ceiling is now concrete:** we have located where the mechanism most likely is, ruled out the classical receptors, and shown that most cochlear candidates and the ion-channel hypotheses are beyond what structure/affinity prediction can decide. Confirmation needs cochlear electrophysiology and a wet collaborator.

5. Honest limitations across the program

- **The DiPT 5-HT_{2A/2C} coverage gap is real and unclosed:** the anchor on the DiPT side of every affinity comparison is inferred, not measured.
 - **Spatial ADME is a proxy** (receptor-expression maps plus computed penetrance), not true regional drug concentration.
 - **Docking on stripped single-chain receptors was noisy** (2.6 to 25 Å amine–D3.32 spread), and pocket-cropping made it worse, not better.
 - **The Boltz benchmark's active fraction (37%) and weak-binder "inactive" band make its absolute AUROC optimistic.**
 - **The cochlear scRNA-seq is mouse P7**, and IHC-vs-OHC subclustering used marker-guided methods that a full annotation would sharpen; human orthology and druggable-pocket existence must be checked per target.
 - **No FEP was run.** Section 3.2 is a design, and its $\Delta\Delta G$ target window is provisional pending measured K_i values.
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6. Complete artifact index

Reports (PDF + markdown) - Original integrated report: `report.pdf` - DiPT ideation / discovery report: `dipt_ideation_report.pdf`, `discovery_program.pdf` - Compute-execution report: `compute_report.pdf` - Phase 2 off-target report: `phase2_report.pdf` - Next-step recommendation (cochlear): `next_step_auditory.pdf` - DPT→DiPT FEP design: `fep_dpt_dipt_design.pdf` - Literature review (33 refs): `literature_review.md`, `references.csv`

Datasets (CSV) - Expanded 1,667-compound landscape: `expanded_compound_dataset.csv` - Boltz co-fold affinities / DiffDock poses / validation pairs: `boltz_affinity_results.csv`, `diffdock_results.csv`, `boltz_validation_pairs.csv` - 100-compound 5-HT_{2A} benchmark — `bench100_results.csv` - Phase 2 profile matrix + results: `phase2_prob_matrix.csv`, `phase2_offtarget_results.csv` - IHC-GPCR target shortlist: `ihc_gpcr_target_shortlist.csv` - Off-target scrape + matrix: `offtarget_bioactivity_scrape.csv`, `offtarget_profile_matrix.csv` - GPCRdb contact comparison + benchmark summary: `gpcrdb_LSD_contact_comparison.csv`, `gpcrdb_benchmark_summary.csv` - Integration master table, ADME descriptors, brain expression: `integration_master_table.csv`, `adme_descriptors.csv`, `receptor_brain_expression.csv`

Figures (PNG): affinity fingerprint, ADME/CNS, brain expression, docking contacts, GPCRdb validation, DiPT focus, landscape, selectivity, SAR, compute results, benchmark, Phase 2 off-target, IHC-GPCR tractability, FEP atom mapping.